Math **163** Name (PLEASE PRINT) Captain Yellowbeard

Exam **2C** **2024** Fall/Spring/**Summer I**/Summer II Score _____

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	4	1	1							6	
1b	3	3								6	
1c											
2a	6									6	
2b	6									6	
2c											
3a	1	1	4							6	
3b	1	1								2	
3c											
								Total		32	

Instructions:

Your J-number = 00407205

Keep every 0 in your J-number and replace every non-zero number in your J-number by 1 to get

Your (truth-value) **T-number**:= 00101101

Define $\forall k \in 1..8$,

T_k := the k[th] entry in your **truth-value J-number** counting from the left. Therefore,

$T_1 := 0$

$T_2 := 0$

$T_3 := 1$

$T_4 := 0$

$T_5 := 1$

$T_6 := 1$

$T_7 := 0$

$T_8 := 1$

Next construct an expression $G(w, x, y, z)$ as follows:

$\alpha(T_1, T_2, T_3, T_4)$

$= (T_1)(w) + (T_2)(x') + (T_3)(y) + (T_4)(z')$

$= (0)(w) + (0)(x') + (1)(y) + (0)(z')$

$= y$

$\beta(T_3, T_4, T_5, T_6)$

$= \left(\left((T_3)w\right)'\right)\left(\left((T_4)x\right)'\right)\left(\left((T_5)y\right)'\right)\left(\left((T_6)z\right)'\right)'$

$= \left(\left((1)\omega\right)'\right)\left(\left((0)x\right)'\right)\left(\left((1)y\right)'\right)\left(\left((1)z\right)'\right)'$

$= \left((\omega')(1)(y')(z')\right)'$

$= \omega + y + z$

$\gamma(T_5, T_6, T_7, T_8)$

$= (T_5)(w') + (T_6)(x) + (T_7)(y') + (T_8)(z)$

$= (1)(\omega') + (1)(x) + (0)(y') + (1)(z)$

$= \omega' + x + z$

=

$$G(w, x, y, z)$$

$$= \quad \big(\alpha(T_1, T_2, T_3, T_4)\big)\big(\beta(T_3, T_4, T_5, T_6)\big)\big(\gamma(T_5, T_6, T_7, T_8)\big)$$

$$= \quad (y)(w + y + z)(w' + x + z)$$

=

6 (1a) Find $\varphi(p,q,r)$ in terms of: p,q and r if $TT\big(\varphi(p,q,r)\big)$ is given below. Find corresponding expressions $F(A,B,C)$ in $\underline{\underline{Sets}}$ and $E(x,y,z)$ in $\underline{\underline{BA}}$ on the facing page and record the answers on this page.

p	q	r	$\varphi(p,q,r)$	φ_1	φ_2	φ_3	φ_4	φ_5	φ_6	φ_7	φ_8
0	0	0	$T_1 = 0$	0	0	0	0	0	0	0	0
0	0	1	$T_2 = 0$	0	0	0	0	0	0	0	0
0	1	0	$T_3 = 1$	0	0	0	0	0	1	0	0
0	1	1	$T_4 = 1$	0	0	0	0	1	0	0	0
1	0	0	$T_5 = 1$	0	0	0	1	0	0	0	0
1	0	1	$T_6 = 1$	0	0	1	0	0	0	0	0
1	1	0	$T_7 = 1$	0	1	0	0	0	0	0	0
1	1	1	$T_8 = 1$	1	0	0	0	0	0	0	0

4(i) $\varphi(p,q,r)$

$= (p \wedge q \wedge r) \vee (p \wedge q \wedge (\neg r)) \vee (p \wedge (\neg q) \wedge r) \vee (p \wedge (\neg q) \wedge (\neg r)) \vee ((\neg p) \wedge q \wedge r) \vee ((\neg p) \wedge q \wedge (\neg r))$

1 (ii) $F(A,B,C)$

$= (A \cap B \cap C) \cup (A \cap B \cap (C^c)) \cup (A \cap (B^c) \cap C) \cup (A \cap (B^c) \cap (C^c)) \cup ((A^c) \cap B \cap C) \cup ((A^c) \cap B \cap (C^c))$

1 (iii) $E(x,y,z)$

$= xyz + xyz' + xy'z + xy'z' + x'yz + x'yz'$

$$
\begin{aligned}
& \varphi(p,q,r) \\
= \; & \varphi_1 \vee \varphi_2 \vee \varphi_3 \vee \varphi_4 \vee \varphi_5 \vee \varphi_6 \\
= \; & (p \wedge q \wedge r) \\
& \vee \; (p \wedge q \wedge (\neg r)) \\
& \vee \; (p \wedge (\neg q) \wedge r) \\
& \vee \; (p \wedge (\neg q) \wedge (\neg r)) \\
& \vee \; ((\neg p) \wedge q \wedge r) \\
& \vee \; ((\neg p) \wedge q \wedge (\neg r))
\end{aligned}
$$

$$
\begin{aligned}
& F(A,B,C) \\
= \; & (A \cap B \cap C) \cup (A \cap B \cap (C^c)) \cup (A \cap (B^c) \cap C) \cup (A \cap (B^c) \cap (C^c)) \cup ((A^c) \cap B \cap C) \\
& \cup ((A^c) \cap B \cap (C^c))
\end{aligned}
$$

$$
\begin{aligned}
& E(x,y,z) \\
= \; & xyz + xyz' + xy'z + xy'z' + x'yz + x'yz'
\end{aligned}
$$

6 (1b) Use the consensus method to find the following for your answer from (1a):

$$E(x,y,z) \quad = \quad xyz + xyz' + xy'z + xy'z' + x'yz + x'yz'$$

3 (i) the prime implicants of E: $PI\ (E) = \{ x, y \}$

3 (i) the minimal sum-of-products form: $MSOP\ (E) = x + y$

1b.i.0

$$
\begin{aligned}
& E \\
= \quad & \underline{xyz} + \underline{xyz'} + xy'z + xy'z' + x'yz + x'yz' && Cns(xyz, xyz') = xy \\
= \quad & \underline{xyz} + \underline{xyz'} + xy'z + xy'z' + x'yz + x'yz' + \underline{xy} && (xy \subseteq xyz,\ xy \subseteq xyz') \\
= \quad & \underline{xy'z} + \underline{xy'z'} + x'yz + x'yz' + xy && Cns(xy'z, xy'z') = xy' \\
= \quad & \underline{xy'z} + \underline{xy'z'} + x'yz + x'yz' + xy + \underline{xy'} && (xy' \subseteq xy'z,\ xy \subseteq xy'z') \\
= \quad & \underline{x'yz} + \underline{x'yz'} + xy + xy' && Cns(x'yz, x'yz') = x'y \\
= \quad & \underline{x'yz} + \underline{x'yz'} + xy + xy' + \underline{x'y} && (x'y \subseteq x'yz,\ x'y \subseteq x'yz') \\
= \quad & \underline{xy} + \underline{xy'} + x'y && Cns(xy, xy') = x \\
= \quad & \underline{xy} + \underline{xy'} + x'y + \underline{x} && (x \subseteq xy,\ x \subseteq xy') \\
= \quad & \underline{x'y} + \underline{x} && Cns(x'y, x) = y \\
= \quad & \underline{x'y} + x + \underline{y} && (y \subseteq x'y) \\
= \quad & x + y
\end{aligned}
$$

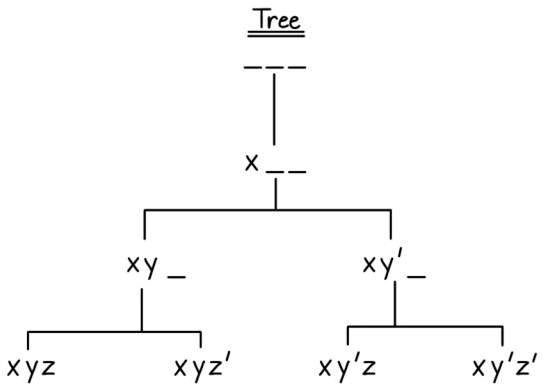
1b.i.1

$$\frac{(x \not\subseteq y)\ \ (y \not\subseteq x)\ \ \cancel{Cns}(x, y)}{PI(E) = \{x, y\}}$$

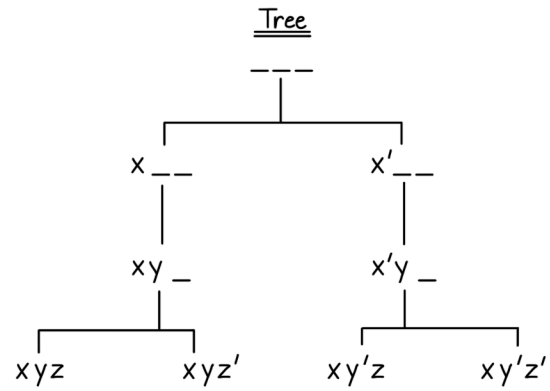
1b.ii.0

$$
\begin{array}{ccccccccccccc}
E & & & & & & & & & & & & \\
= & x & = & x(1)(1) & = & x(y + y')(z + z') & = & xyz(11) + xyz'(21) + xy'z + xy'z' & = & xy(11) + xy' & = & x & = MSOP(E) \\
 & \big| + y & & \big| + (1)y(1) & & \big| + (x + x')y(z + z') & & \big| + xyz(12) + xyz'(22) + x'yz + x'yz' & & \big| + xy(12) + x'y & & \big| + y & \big|
\end{array}
$$

1b.ii.1



1b.ii.2



6 (2a) For the expression from page 605 below:

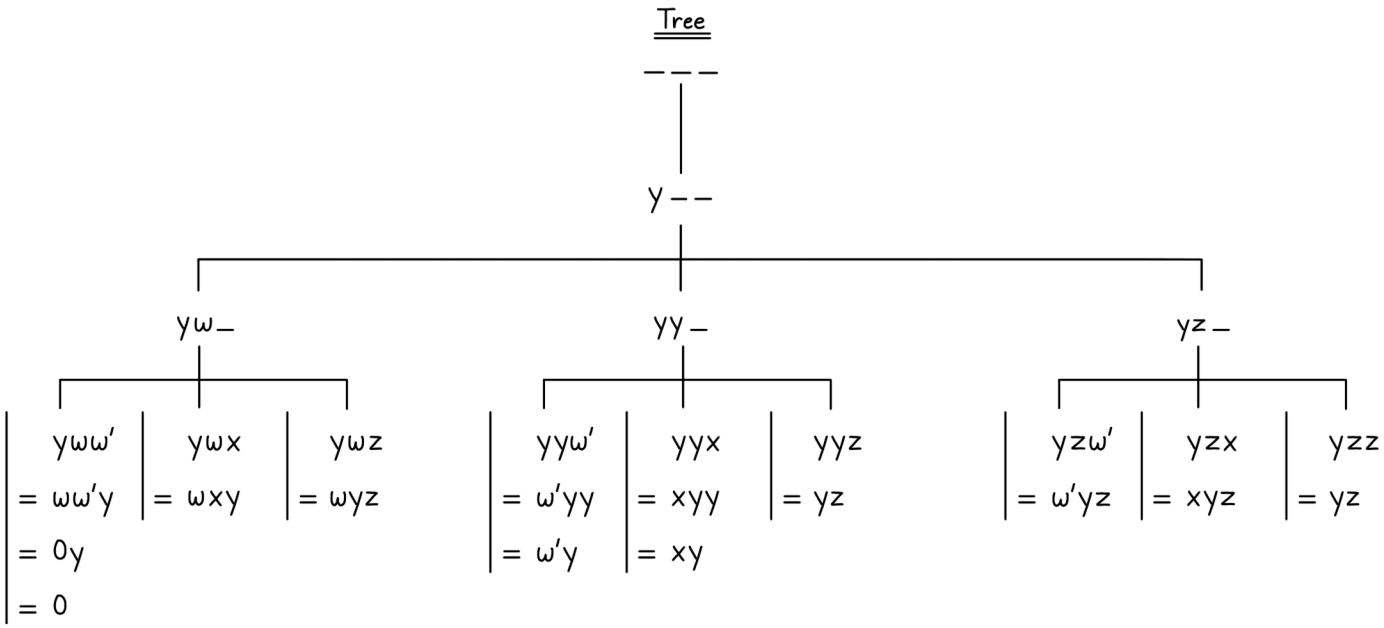
$$G(w,x,y,z) = (y)(w + y + z)(w' + x + z)$$

find a sum-of-products form, $SOP(G)$

$$SOP(G) = w'y + xy + yz$$

Show the tree on the facing side.

2a.0



2a.1

$$
\begin{aligned}
& G(\omega,x,y,z) \\
= \;& 0 + \omega xy + \omega yz + \omega' y + xy + yz + \omega' yz + xyz + yz \\
= \;& \underline{\omega xy} + \underline{\omega yz} + \underline{\omega' y} + \underline{xy} + \underline{yz} + \underline{\omega' yz} + \underline{xyz} \\
= \;& \omega' y + xy + yz
\end{aligned}
$$

$(xy \subseteq \omega xy,\ \ xy \subseteq xyz,\ \ yz \subseteq \omega yz,\ \ yz \subseteq \omega' yz)$

2a.2

$\omega' y \not\subseteq xy \quad \omega' y \not\subseteq yz$

$xy \not\subseteq \omega' y \quad xy \not\subseteq yz$

$yz \not\subseteq \omega' y \quad yz \not\subseteq xy$

2a.3

$$
\begin{aligned}
& G(\omega,x,y,z) \\
= \;& \omega' y + xy + yz \\
= \;& \mathrm{SOP}(G)
\end{aligned}
$$

6120

6 (2b) For the expression below (**which is the same as in (2a)**):

$$G(w,x,y,z) = (y)(w + y + z)(w' + x + z)$$

find the complete sum-of-products form, $CSOP(G)$

$$\begin{aligned} CSOP(G) \quad = \quad & w'xyz \\ & + w'xyz' \\ & + w'x'yz \\ & + w'x'yz' \\ & + wxyz \\ & + wxyz' \\ & + wx'yz \end{aligned}$$

26.0

SOP(G)

$= \omega'y + xy + yz$

$= \omega'(1)y(1) + (1)xy(1) + (1)(1)yz$

$= \omega'(x+x')y(z+z') + (\omega+\omega')xy(z+z') + (\omega+\omega')(x+x')yz$

$= \omega'xyz(11) + \omega'xyz'(21) + \omega'x'yz(31) + \omega'x'yz' + \omega xyz(41) + \omega xyz' + \omega'xyz(12) + \omega'xyz'(22) + \omega xyz(42) + \omega x'yz + \omega'xyz(13) + \omega'x'yz(32)$

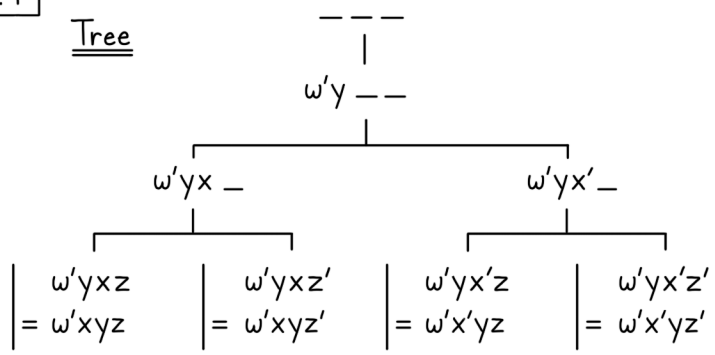
$= \omega'xyz + \omega'xyz' + \omega'x'yz + \omega'x'yz' + \omega xyz + \omega xyz' + \omega x'yz$

$= \omega'xyz + \omega'xyz' + \omega'x'yz + \omega'x'yz' + \omega xyz + \omega xyz' + \omega x'yz$

$=$ CSOP(G)

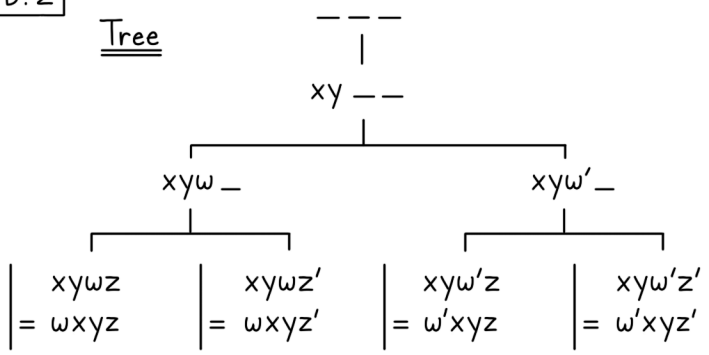
26.1

Tree



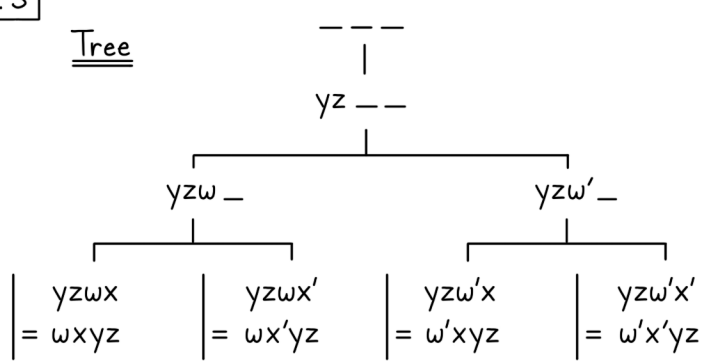
26.2

Tree



26.3

Tree



6140

5 (2c) For the expression below **which is the same as in (2a) and (2b)**:

$$G(w,x,y,z) = (y)(w + y + z)(w' + x + z)$$

use your answer from (2a) and (2b) to find the following:

$PI(G)$ and $MSOP\ (G)$

Answer from (2a) $SOP(G) =$ w′y + xy + yz

Answer from (2b) $CSOP\ (G) =$ w′xyz + w′xyz′ + w′x′yz + w′x′yz′
+ wxyz + wxyz′ + wx′yz

2 (i) $PI(G) = \{w'y, xy, yz\}$

3 (ii) $MSOP\ (G) =$ w′y + xy + yz

2c.i.0

$$\frac{\begin{array}{c}\text{SOP}(G) \\ = w'y + xy + yz\end{array}}{\text{PI}(G) = \{w'y, xy, yz\}}$$

2c.i.1

G

$= \underline{w'xyz} + \underline{w'xyz'} + w'x'yz + w'x'yz' + wxyz + wxyz' + wx'yz$ — $\text{Cns}(w'xyz, w'xyz') = w'xy$

$= \underline{w'xyz} + \underline{w'xyz'} + w'x'yz + w'x'yz' + wxyz + wxyz' + wx'yz + \underline{w'xy}$ — $(w'xy \subseteq w'xyz, w'xyz')$

$= \underline{w'x'yz} + \underline{w'x'yz'} + wxyz + wxyz' + wx'yz + w'xy$ — $\text{Cns}(w'x'yz, w'x'yz') = w'x'y$

$= \underline{w'x'yz} + \underline{w'x'yz'} + wxyz + wxyz' + wx'yz + w'xy + \underline{w'x'y}$ — $(w'x'y \subseteq w'x'yz, w'x'yz')$

$= \underline{wxyz} + \underline{wxyz'} + wx'yz + w'xy + w'x'y$ — $\text{Cns}(wxyz, wxyz') = wxy$

$= \underline{wxyz} + \underline{wxyz'} + wx'yz + w'xy + w'x'y + \underline{wxy}$ — $(wxy \subseteq wxyz, wxyz')$

$= wx'yz + \underline{w'xy} + \underline{w'x'y} + wxy$ — $\text{Cns}(w'xy, w'x'y) = w'y$

$= wx'yz + \underline{w'xy} + \underline{w'x'y} + wxy + \underline{w'y}$ — $(w'y \subseteq w'xy, w'x'y)$

$= \underline{wx'yz} + \underline{wxy} + w'y$ — $\text{Cns}(wx'yz, wxy) = wyz$

$= \underline{wx'yz} + wxy + w'y + \underline{wyz}$ — $(wyz \subseteq wx'yz)$

$= \underline{wxy} + \underline{w'y} + wyz$ — $\text{Cns}(wxy, w'y) = xy$

$= \underline{wxy} + w'y + wyz + \underline{xy}$ — $(xy \subseteq wxy)$

$= \underline{w'y} + \underline{wyz} + xy$ — $\text{Cns}(w'y, wyz) = yz$

$= w'y + \underline{wyz} + xy + \underline{yz}$ — $(yz \subseteq wyz)$

$= w'y + xy + yz$

2c.i.2

$w'y \not\subseteq xy$	$w'y \not\subseteq yz$	~~Cns~~$(w'y, xy)$
$xy \not\subseteq w'y$	$xy \not\subseteq yz$	~~Cns~~$(w'y, yz)$
$yz \not\subseteq w'y$	$yz \not\subseteq xy$	~~Cns~~(xy, yz)

2c.i.3

$\text{PI}(G) = \{w'y, xy, yz\}$

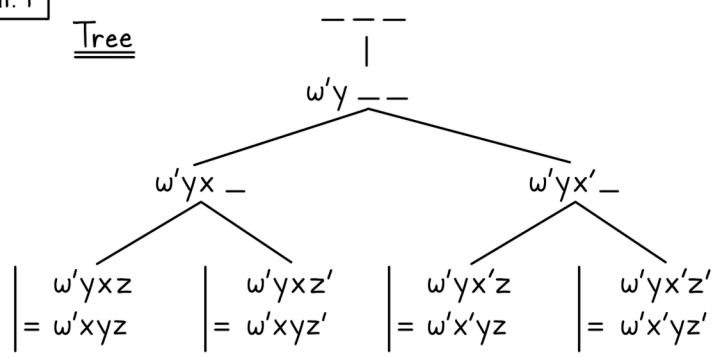
2c.ii.0

G

$$= \begin{array}{|l} \omega'y \\ + xy \\ + yz \end{array} = \begin{array}{|l} \omega'(1)y(1) \\ + (1)xy(1) \\ + (1)(1)yz \end{array} = \begin{array}{|l} \omega'(x+x')y(z+z') \\ + (\omega+\omega')xy(z+z') \\ + (\omega+\omega')(x+x')yz \end{array} = \begin{array}{|l} \omega'xyz(11) + \omega'xyz'(21) + \omega'x'yz(31) + \omega'x'yz' \\ + \omega xyz(41) + \omega xyz' + \omega'xyz(12) + \omega'xyz'(22) \\ + \omega xyz(42) + \omega x'yz + \omega'xyz(13) + \omega'x'yz(32) \end{array} = \begin{array}{|l} \omega'y \\ + xy \\ + yz \end{array} = MSOP(G)$$

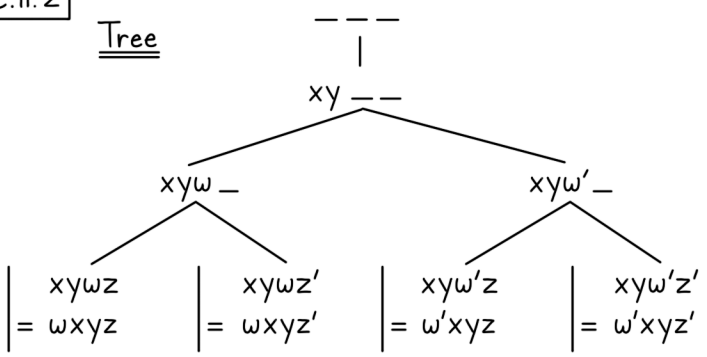
2c.ii.1

Tree



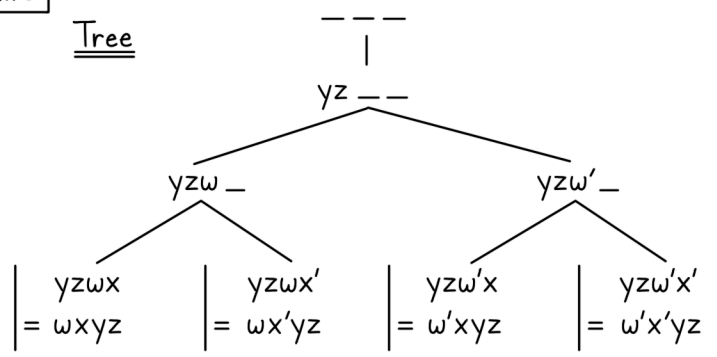
2c.ii.2

Tree



2c.ii.3

Tree



6 (3a) We define:

$A :=\ SOP\big(G(w,x,y,z)\big) =$ w'y + xy + yz

$B\ =\ MSOP\big(G(w,x,y,z)\big) =$ w'y + xy + yz

Copy your answers from (1a)

Compute:

1 (i) $A_S\ = 3$ $A_L\ = 6$ $B_S\ = 3$ $B_L\ = 6$

1 (ii) On the basis of your answers to (i), circle the correct choice below.

$Smplr\ (A,B)$ $Smplr(B,A)$ (Neither)

Why?

4 (iii) Write down the version of $G(w,x,y,z)$ in PC below and find its **formation-tree on the facing side**.

Version of $G(w,x,y,z)$in PC

= $(r) \wedge (p \vee r \vee s) \wedge ((\neg p) \vee q \vee s)$

3a.i.0

$$\begin{aligned} & A_S \\ = \ & v(Smnd(A)) \\ = \ & v(\{\omega' y, xy, yz\}) \\ = \ & 3 \end{aligned}$$

3a.i.1

$$\begin{aligned} & A_L \\ = \ & v(MSLtr(A)) \\ = \ & v(\{\omega' : 1, x : 1, y : 3, z : 1\}) \\ = \ & 6 \end{aligned}$$

3a.i.2

$$\begin{aligned} & B_S \\ = \ & v(Smnd(B)) \\ = \ & v(\{\omega' y, xy, yz\}) \\ = \ & 3 \end{aligned}$$

3a.i.3

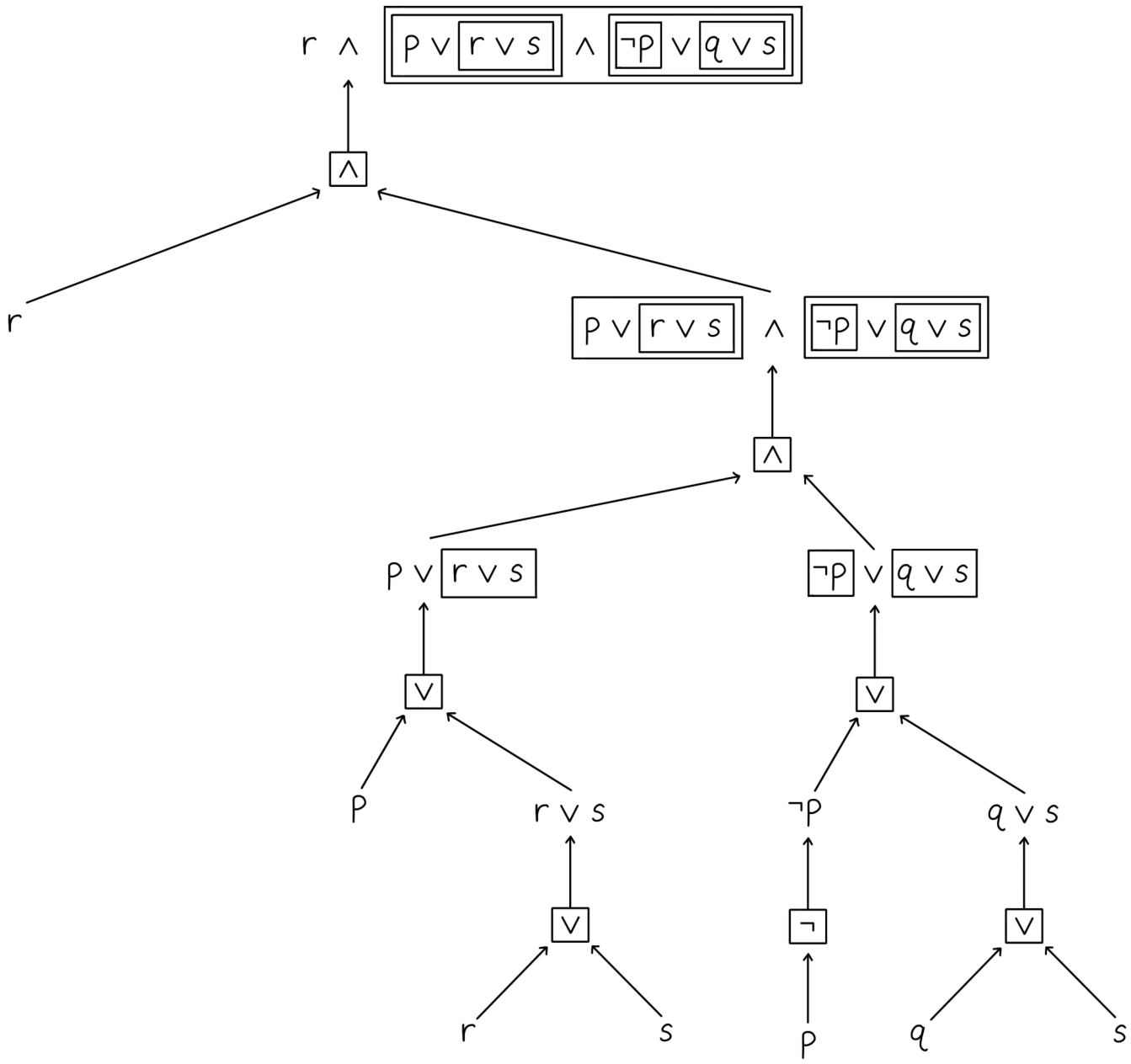
$$\begin{aligned} & B_L \\ = \ & v(MSLtr(B)) \\ = \ & v(\{\omega' : 1, x : 1, y : 3, z : 1\}) \\ = \ & 6 \end{aligned}$$

3a.ii

$$\frac{\left(\neg\left(\left(A_L = 6 < 6 = B_L\right) \wedge \left(A_S = 3 \leq 3 = B_S\right)\right)\right) \quad \left(\neg\left(\left(B_L = 6 < 6 = A_L\right) \wedge \left(B_S = 3 \leq 3 = A_S\right)\right)\right)}{\cancel{Smptr}(A, B) \wedge \cancel{Smptr}(B, A)}$$

3a.iii.0

$$\frac{G(\omega, x, y, z) = (y)(\omega + y + z)(\omega' + x + z)}{\begin{aligned} & \Psi(p, q, r, s) \\ = \ & (r) \wedge (p \vee r \vee s) \wedge ((\neg p) \vee q \vee s) \end{aligned}}$$


r ∧ p ∨ r ∨ s ∧ ¬p ∨ q ∨ s
∧
r
p ∨ r ∨ s ∧ ¬p ∨ q ∨ s
∧
p ∨ r ∨ s
¬p ∨ q ∨ s
∨
∨
p
r ∨ s
¬p
q ∨ s
∨
¬
∨
r
s
p
q
s

2 (3b) Find the versions: $H(A,B,C,D)$ of $G(w,x,y,z)$

and $K(A,B,C,D)$ of $MSOP\big(G(w,x,y,z)\big)$

in Sets.

1 (i) $H(A,B,C,D)$

= $(C)\cap(A\cup C\cup D)\cap((A)^c\cup B\cup D)$

1 (i) $K(A,B,C,D)$

= $((A)^c\cap C)\cup(B\cap C)\cup(C\cap D)$

3b.i

$$G(w,x,y,z) = (y)(w+y+z)(w'+x+z)$$

$$H(A,B,C,D) = (C) \cap (A \cup C \cup D) \cap ((A)^c \cup B \cup D)$$

3b.ii

$$MSOP(G(w,x,y,z)) = w'y + xy + yz$$

$$K(A,B,C,D) = ((A)^c \cap C) \cup (B \cap C) \cup (C \cap D)$$